

Github Link: https://github.com/lisha080/CSE464_2026_lseekalr

mvn package result:

```
[INFO] -----
[INFO]  T E S T S
[INFO] -----
[INFO] Running CSE464ProjectP1.GraphTest
[INFO] Tests run: 5, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.028 s -
- in CSE464ProjectP1.GraphTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 5, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO] --- jar:3.4.1:jar (default-jar) @ CSE464_2026_lseekalr ---
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 0.901 s
[INFO] Finished at: 2026-03-06T20:31:54-07:00
[INFO] -----
(base) lisharamamurthy@MacBook-Pro-7 CSE464-P1 %
```

Commits History:

https://github.com/lisha080/CSE464_2026_lseekalr/commits/main/src/main/java/CSE464ProjectP1

- Initial Commit:
https://github.com/lisha080/CSE464_2026_lseekalr/commit/bab0261e3116029086a25d7cfa4f03337a56d353
- Feature 1 Commit:
https://github.com/lisha080/CSE464_2026_lseekalr/commit/2cc3e32bcd94306fa193a10aa8ce145f6a70bb70
- Feature 2 Commit:
https://github.com/lisha080/CSE464_2026_lseekalr/commit/edd73a6975a99f149829d43c765939ad1ee0ee07
- Feature 3 Commit:
https://github.com/lisha080/CSE464_2026_lseekalr/commit/5bb2bb0bd40b78c811b8dd66e9ec9b10b3a57dbb
- Feature 4 Commit:
https://github.com/lisha080/CSE464_2026_lseekalr/commit/22be1a59eba1e49b56e427e150bef8bbd0574b5a
- Unit Tests Commit:
https://github.com/lisha080/CSE464_2026_lseekalr/commit/042057c7a73fe2e46a8eac0b1276385b2f9c3c3c

Input and expected outputs:

input.dot file:

```
⋮ input.dot ×
1      digraph {
2      a -> b;
3      b -> c;
4      c -> d;
5      d -> a;
6      a -> e;
7      e -> f;
8      e -> g;
9      f -> h;
10     g -> h;
11     }
```

Feature 1 input + output:

```
Graph graph = new Graph();
graph.parseGraph( filepath: "input.dot");
System.out.println(graph);
```

```
Number of nodes: 8
Node labels:
a
b
c
d
e
f
g
h
Number of edges: 9
Edges:
a -> b
b -> c
c -> d
d -> a
a -> e
e -> f
e -> g
f -> h
g -> h
```

Feature 2 input + output:

```
graph.addNode( label: "x");  
graph.addNodes(new String[]{"y","z"});  
System.out.println(graph);
```

Number of nodes: 11

Node labels:

a
b
c
d
e
f
g
h
x
y
z

Number of edges: 9

Edges:

a -> b
b -> c
c -> d
d -> a
a -> e
e -> f
e -> g
f -> h
g -> h

Feature 3 input + output:

```
graph.addEdge( srcLabel: "x", dstLabel: "a");  
graph.addEdge( srcLabel: "a", dstLabel: "b");  
System.out.println(graph);
```

Number of nodes: 11

Node labels:

a
b
c
d
e
f
g
h
x
y
z

Number of edges: 10

Edges:

a -> b
b -> c
c -> d
d -> a
a -> e
e -> f
e -> g
f -> h
g -> h
x -> a

Feature 4 input + output:

```
graph.outputDOTGraph(path: "output.dot");  
graph.outputGraphics(path: "output.png", format: "png");  
System.out.println(graph);
```

Number of nodes: 11

Node labels:

a

b

c

d

e

f

g

h

x

y

z

Number of edges: 10

Edges:

a -> b

b -> c

c -> d

d -> a

a -> e

e -> f

e -> g

f -> h

g -> h

x -> a

output.dot file

```
1 digraph {
2   a -> b;
3   b -> c;
4   c -> d;
5   d -> a;
6   a -> e;
7   e -> f;
8   e -> g;
9   f -> h;
10  g -> h;
11  x -> a;
12 }
```

output.png

